Example 9.46

Evaluate $\int_0^{\infty} \frac{x^n}{n^x} dx$, where n is a positive integer ≥ 2.

Solution

Using the formula $n = e^{\log_e n}$, we get

$$I = \int_0^{\infty} \frac{x^n}{n^x} dx = \int_0^{\infty} n^{-x} x^n dx = \int_0^{\infty} \left(e^{\log n}\right)^{-x} x^n dx = \int_0^{\infty} e^{-x \log n} x^n dx .$$

Using the substitution $u = x \log n$, we get $dx = \frac{du}{\log n}$.

When $x = 0$, we get $u = 0$. When $x = \infty$, we get $u = \infty$.

$$\therefore I = \int_0^{\infty} e^{-u} \left(\frac{u}{\log n}\right)^n \frac{du}{\log n}$$

$$= \frac{1}{(\log n)^{n+1}} \int_0^{\infty} e^{-u} u^{(n+1)-1} du = \frac{\Gamma(n+1)}{(\log n)^{n+1}} = \frac{n!}{(\log n)^{n+1}} .$$

EXERCISE 9.7

Evaluate the following

1. (i) $\int_0^{\infty} x^5 \, e^{-3x} \, dx$ (ii) $\int_0^{\frac{\pi}{2}} \frac{e^{-\tan x}}{\cos^6 x} dx$

2. If $\int_0^{\infty} e^{-\alpha x^2} x^3 dx = 32$, $\alpha > 0$, find α

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9.8 Evaluation of a Bounded Plane Area by Integration

In the beginning of this chapter, we have already introduced definite integral by a geometrical approach. In that approach, we have noted that, whenever the integrand of the definite integral is non-negative, the definite integral yields the geometrical area. In the present section, we apply the approach for finding areas of plane regions bounded by plane curves.

9.8.1 Area of the region bounded by a curve, x – axis and the lines $x = a$ and $x = b$.

Case (i)

Let $y = f(x)$, $a \leq x \leq b$ be the equation of the portion of the continuous curve that lies above the $x-$axis (that is, the portion lies either in the first quadrant or in the second quadrant) between the lines $x = a$ and $x = b$. See Fig.9.8. Then, $y \geq 0$ for every point of the portion of the curve. Consider the region bounded by the curve, $x-$axis, the ordinates $x = a$ and $x = b$. It is important to note that y does not change its sign in the region. Then, the area A of the region is found as follows:

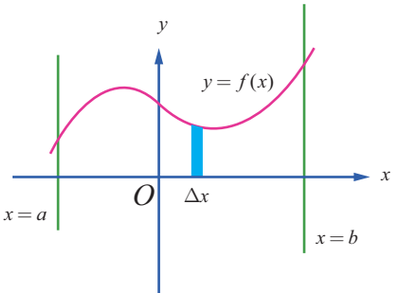


Fig. 9.8

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